

Speculative Trading in the Gold Market

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Abstract

In this paper we use a recently developed econometric test to identify bubble-like price behaviour in the gold market. We find that the price of gold followed an explosive price process between 2002 and 2012 and furthermore exhibited super-exponential growth between 2002 and 2008, indicating excessive speculative trading and exuberance in the gold market. We also provide a theoretical foundation for such bubble tests based on a behavioural model in which chartists can cause episodes of explosive price dynamics. The identification strategy yields economically intuitive results and is a simple alternative to using more complex estimation techniques commonly used in the heterogenous agents literature.

JEL classification: C22; C51; G02

Keywords: speculation; gold; bubble; heterogeneous agents; chartists; explosive process

*The first version of this paper was entitled “A Gold Bubble?” and reported evidence for bubble-like characteristics in gold prices between 2002 and 2012. This version of the paper confirms this result with an updated dataset including the “correction” in the price of gold in 2013.

Combating asset price bubbles ... requires us to be successful in both identifying the incipient bubble and in developing and implementing a response that will limit bubble growth and avert a destructive asset price crash.

—William C Dudley, President and CEO of the Federal Reserve Bank of New York at the Economic Club of New York, NYC, 7 April 2010.

The ultimate asset bubble is gold.

—George Soros, The World Economic Forum 2010, Davos, Switzerland.¹

Asset price bubbles have presented an important (and difficult) challenge to researchers and policymakers for centuries.² Yet there is still not a widespread consensus on the definition of a bubble or the causes of such price dynamics. This paper aims to enhance the understanding of the causes of bubble-like price behaviour and to address the empirical question of whether or not the price of gold has exhibited such dynamics.³

Recent renewed interest in asset price bubbles, precipitated by the subprime crisis or global economic and financial crisis in 2008, has led to the invention of new tools to identify asset price bubbles. More specifically, bubbles can now be detected from their *time-series* characteristics using techniques developed in a series of works by Peter C. B. Phillips and co-authors.⁴ These tools allow us to detect early warning signs of potential bubble-like behaviour, meaning that it is no longer

¹See <http://www.telegraph.co.uk/finance/financetopics/davos/7085504/Davos-2010-George-Soros-warns-gold-is-now-the-ultimate-bubble.html>. At this time Soros also massively reduced his stake in gold ETFs. Gold subsequently recorded annual gains in 2010 and 2011, reaching a high of US\$1,920.30 per ounce in September 2011.

²For a detailed history see Kindleberger (2005) or more recently Scherbina and Schlusche (2013).

³We focus primarily on the detection of bubble-like behaviour, however there are many related questions regarding the appropriate policy response if such behaviour is indeed detected. In this regard, the heterogeneous agents foundation for the bubble tests presented in Section 2 may be useful.

⁴See, for example, Phillips and Magdalinos (2007), Phillips et al. (2011), Phillips and Yu (2011) and Phillips et al. (2012). Technically, the method relies on recursive regressions coupled with sequential right-sided unit-root tests.

necessary to wait for a market correction in order to identify bubble-like behaviour.⁵ Such an ability to detect bubbles prior to their collapse is crucial if policies are to be implemented in a timely manner in order to reduce the severity of the inevitable price correction; and the resulting impact on financial stability and the real economy.

In relation to gold, only a cursory inspection of the time series of gold prices over the last 45 years (see Figure 1) begs the question as to whether or not gold has been experiencing bubble behaviour since the turn of the century—making gold an ideal candidate for the application of these newly developed tests. The aim of this paper, therefore, is to employ the econometric techniques developed by Phillips and co-authors to test if, and indeed when, a ‘bubble’ has emerged (and collapsed) in the market for gold. The ability to tell whether gold is or is not unduly influenced by speculative trading behaviour has important implications given that gold is viewed by many as a store of value and a safe haven (see Baur and Lucey, 2010). If the price of gold is inflated and exhibits bubble-like characteristics the safe-haven property can be destroyed and with it the stabilizing effect of the existence of such an asset on the financial system.⁶

When considering the NASDAQ equity market Phillips et al. (2011) demonstrate the existence of a bubble during the 1990s by identifying explosive behaviour in the dynamics of the asset price but not in the dynamics of the fundamental value.⁷ However, when considering commodity markets, which do not fit into the theoretical framework of rational bubbles (since they provide no future cash flows), Phillips and

⁵Gürkaynak (2008) provides a detailed survey of the advances in the econometric detection of asset price bubbles up to 2008. Other notable contributions post 2008 include Al-Anaswah and Wilfling (2011), Gutierrez (2011), Frömmel and Kruse (2012), Homm and Breitung (2012), Asako and Liu (2013) and Corsi and Sornette (2014).

⁶Baur and Glover (2012) study the mechanisms through which investors could potentially destroy the safe-haven property of gold and similar safe-haven assets.

⁷Cuñado et al. (2005) also test for bubbles in the NASDAQ using a different, fractionally integrated, approach; again finding evidence of the existence of a bubble during this period.

Yu (2011) interpret explosiveness in price as sufficient evidence of speculator induced bubble-like behaviour (page 461). One obvious advantage of such an interpretation is that the fundamental value need not be explicitly modelled. Thus, for those assets whose fundamental value is opaque, this interpretation removes any errors in the accuracy of such tests due to the misspecification of the fundamental value. Given the distinct lack of a credible (consensus) fundamental value for gold we also adopt the interpretation of Phillips and Yu (2011) in the present paper.⁸ Hence detection of explosive price behaviour in gold prices should be taken as a warning sign of potential bubble-like price dynamics.⁹

In addition, similarly to Phillips et al. (2011) (see also Diba and Grossman, 1988) who demonstrate that a *rational* bubble leads to explosive price behaviour, we justify an explosive price process from an alternative behavioural, *heterogenous agents*, foundation. Therefore, instead of merely detecting bubble-like behaviour in the gold market, our behavioural foundation for the dynamics estimated in the bubble test allows us to glean information about the activities of speculative traders in the gold market over time. These results indicate that the number and impact of speculative traders varies through time and that in periods where specific groups of traders are more active the price dynamics change from a mean-reverting process (consistent with the real supply and demand for gold) to an explosive process.

⁸Indeed, articles in the Financial Times, The Economist and the Wall Street Journal suggest that there is a consensus that the value of gold is *unknown*. Gold is “hard to value” (The Economist, 2013), “has no fundamental value” (Financial Times, 2013), and “can’t be measured in the same way as other financial assets” (Wall Street Journal, 2013).

⁹We define an explosive price process in the (usual) context of an autoregressive model as in Phillips and Yu (2011).

Related literature

The precipitous rise in the price of gold in recent years has not gone unnoticed by the academic community. Other recent studies have also attempted to assess whether the gold market is experiencing a price bubble. These papers, however, use different methodologies and as such reach differing conclusions. Bialkowski et al. (2011) attempt to test for a bubble in the price of gold by estimating the fundamental value of gold using the convenience yield model of Pindyck (1993). While this method produces some interesting results (indicating that a gold bubble does *not* exist) we choose instead to employ a method that needs less assumptions on the knowledge of the fundamental value of gold. Furthermore, in a recent update of their paper, Bialkowski et al. (2012) modify their approach by constructing a more comprehensive fundamental value of gold from a multivariate regression of gold returns on proxies for the various roles that gold plays in the financial system, such as a US dollar hedge, an inflation hedge, a portfolio diversifier, and a safe haven. While this method also produces some interesting results (again indicating no gold bubble) we believe that some of the factors modelled here should not be interpreted as *fundamental* factors but instead as factors that have induced speculative investors into the gold market, leading to bubble-like behaviour. Indeed, we believe that if enough factors were added to the regression model for the fundamental value one would always be able to explain away any bubble-like behaviour of gold prices.

Another recent bubble test for the gold market can be found in Jarrow et al. (2011) who do not use any notion of the fundamental value of gold. These authors also reach the conclusion that the recent rise in the gold price was not a bubble, however their methodology is very different to the one presented in this paper and is based on an analysis of the volatility structure of gold prices; the intuition being

that speculative trading abnormally increases the asset's volatility. It must be noted, however, that these authors only employ one week of tick data from August 2011 to test for bubble behaviour.

Evidence in support of a bubble in the gold market includes Homm and Breitung (2012), who also adopt and modify the framework of Phillips et al. (2011). As an application they consider many asset markets, including only a very limited analysis of gold, finding evidence of bubble-like behaviour in the gold price; detecting significant explosive behaviour (at the 10% level) for the period January 1985 to November 2010. We note that the conclusions of Homm and Breitung (2012) also do not rely on any assumption of a fundamental value of gold, stating that “detecting a change from $I(1)$ to explosive clearly points to excessive speculation”.

In regards to the bubble tests themselves, Phillips et al. (2011) introduced a recursive regression methodology to investigate the explosive characteristics of asset prices. Phillips and Yu (2011) later modified this methodology to test for bubble migration across different asset markets. The latter paper concluded that a bubble in the real estate market, starting in February 2002, migrated selectively into both the commodity market and the bond market following the onset of the subprime crisis in 2007. They suggest that flight-to-quality or perceived safe-haven phenomena were at play during this time. Notably absent in the analysis of Phillips and Yu (2011) is the gold market, which appears to be an ideal candidate for such tests. Our paper fills this gap. We note that our results indicate that explosive behaviour in the gold market may have originated as early as 2002, well before the onset of the subprime crisis.

Finally, an important strand of the extant literature, intimately related to the issue of bubbles in commodity markets such as gold, investigates the role that insti-

tutional investors have played in driving commodity prices in recent years. Since the turn of the century, commodities markets (including gold) have seen the investment of billions of dollars from institutional investors and, as such, commodities now constitute a considerable proportion of many investors' portfolios. As a result of this, it is believed by many that a structural change in the commodities markets has occurred, leading to a process known as 'the financialization of commodities' (see Tang and Xiong, 2012). The origination of the explosive behaviour in gold prices in 2002, identified in the present paper, coincides with the financialization of commodities in general and the introduction of exchange-traded funds on gold in particular (see World Gold Council, 2012). The behavioural foundations for the price dynamics proposed in Section 2 are also in line with this process and provide an alternative theoretical justification for the bubble tests employed.¹⁰

This paper contributes to the literature in two important ways. First, it presents an empirical application of a recently developed bubble-identification framework to an asset of global importance and second, it provides an alternative foundation for such a test based on the heterogeneous agents literature. This paper also demonstrates that there is a rather simple alternative to the estimation of complex switching mechanisms commonly used in the heterogeneous agents literature for identifying periods of speculative trader activity.

The bubble-identification methodology reveals that gold has exhibited bubble-like characteristics and an explosive price behaviour in recent years. Furthermore, our behavioural foundations for the bubble test allow us to interpret the p-value generated from such tests as a simple (and almost natural) proxy for the weight or

¹⁰We note that Arora and Shi (2012) also propose a heterogeneous agent foundation for certain asset price bubble tests. However, these authors refer to tests developed by van Norden (1996) and Brooks and Katsaris (2005)—based on differing bubble regimes—rather than the more recent tests proposed by Phillips et al. (2011) and considered in the present paper.

the importance of speculative traders relative to real demand and supply factors or investors following more stabilizing trading strategies.

The remainder of this paper is structured as follows: Section 1 describes the econometric framework to test for explosive price behaviour and bubbles. Section 2 presents the behavioural foundations for the test. Section 3 describes and discusses the estimation results. Finally, Section 4 summarizes the findings and concludes.

1 Econometric Framework

To detect evidence of explosive behaviour in the price of gold we follow Phillips, Wu, and Yu (2011)—henceforth PWY—and test for a mildly explosive price process using recursive right-sided unit-root tests.¹¹ We use different start dates or initialization dates and can identify periods of statistically significant explosive price behaviour and periods of statistically insignificant explosive price behaviour.¹²

More formally, we apply the augmented Dickey-Fuller (ADF) test and estimate the following autoregressive specification by recursive least squares

$$P_t = \mu + \delta P_{t-1} + \sum_{j=1}^J \phi_j \Delta P_{t-j} + \epsilon_t. \quad (1)$$

where P denote gold prices, $\Delta P_t := P_t - P_{t-1}$ and $\epsilon_t \sim iid(0, \sigma^2)$.

The model is estimated repeatedly using subsets of the sample data incremented by one observation at each iteration. It is a forward recursive estimation strategy since the number of observations increases moving forward in time at each iteration.

¹¹Phillips et al. (2011) state that mildly explosive behaviour may be modelled by an autoregressive process with a root that exceeds unity but that is still in the general vicinity of unity, i.e. the process would involve only mild departures from strict (rational) martingale behaviour in markets.

¹²An explosive process can be thought of as a process whose absolute price changes are higher the higher the level of the price. We will return to this notion of an explosive process in the theoretical model proposed below.

The lag order J is determined following the approach by Campbell and Perron (1991). The null hypothesis is $H_0 : \delta = 1$ and the right-tailed alternative hypothesis is $H_1 : \delta > 1$ which allows for mildly explosive autoregressions.

We use the resulting ADF t-test statistic and the associated p-value calculated for each (recursive) sample up to observation t to analyze whether there are periods in which the price process is mildly explosive and thus consistent with a bubble. Furthermore, we show in the next section that an explosive process is consistent with the influence of chartist traders whilst a non-explosive process is consistent with the influence of real supply and demand factors and investors who do not follow such chartist strategies.

2 A behavioural foundation for the bubble test

The aim of this section is to provide an alternative behavioural foundation for the bubble test of PWY. We demonstrate that there exists a theoretical mapping between the ADF test statistic (and the associated p-values) of such bubble tests and the activity and importance of certain chartist groups commonly used in the heterogeneous agents literature.¹³ We assume the following structural model for the price formation (see, for example, Farmer and Joshi, 2002; He and Westerhoff, 2005)

$$P_{t+1} = P_t + \theta \left(\sum_h D_t^h - S_t \right) + e_t \quad (2)$$

where P_t denotes the gold price, D_t^h the demand of agents of ‘type’ h , and S_t the available supply of the asset. The types of agents active in a given market and their associated asset demands can be wide-ranging and dependent on the particular

¹³For a detailed survey of the history and recent advances in heterogeneous agents models see Hommes (2006), Chiarella et al. (2009), Hommes and Wagener (2009), or more recently Chen et al. (2012).

application. In the present context of the gold market, which is both a real and a financial asset, there is clearly demand for gold in industrial uses, the manufacture of jewelry and as an investment or speculative asset. We therefore choose to model one agent type to represent real consumer demand, denoted as D_t^R . We also explicitly model the demands of two types of speculative or ‘chartist’ behaviour, which we call ‘trend followers’ and ‘threshold traders’, denoted by D_t^{CF} and D_t^{CT} respectively. In a departure from the usual heterogeneous agent models, we do not attempt to explicitly model the influence of any gold market ‘fundamentalists’ since, as we alluded to above, we believe that the lack of consensus on a fundamental value of gold, and the resulting dispersion in agents’ beliefs, would render such agents’ aggregate demand tantamount to noise (and so can be incorporated in the noise term e_t above).

Consistent with He and Westerhoff (2005), amongst others, the real demand and supply of gold is modelled in the following stylized and intuitive way

$$D_t^R = a_R - b_R P_t, \quad S_t = a_S + b_S P_t,$$

where $b_R > 0$ and $b_S > 0$, indicating that the real demand for gold decreases, and the supply of gold increases, as the gold price rises. The chartists’ demand for gold, however, is dependent on their (heterogenous) beliefs about future price movements and is modelled by

$$D_t^h = \sum_{j=1}^{N_t^h} a_j^h \left(\mathbb{E}_t^{h,j}[P_{t+1}] - P_t \right), \quad h = CF, CT$$

where a_j^h are reaction parameters and N_t^h denotes the (potentially time-dependent) number of agents of type h active in the market at time t . The expectations are

taken with respect to each agent's subjective belief. Such beliefs are modelled to be consistent with common trading strategies employed by many traders and for our two groups of chartists they are chosen to be

$$\mathbb{E}_t^{CF,j}[P_{t+1}] = P_t + b_j^{CF}(P_t - P_{t-k_j}), \quad \mathbb{E}_t^{CT,j}[P_{t+1}] = P_t + b_j^{CT}(P_t - c_t^j),$$

where the extrapolation parameters b_j^{CF} and b_j^{CT} , the (possibly time-dependent) threshold c_t^j , and the time window over which the trend followers extrapolate past trends k_j , are all allowed to vary across different agents.

At this stage we pause to note that the threshold traders above could also be interpreted as ‘contrarian fundamentalists’ if we identified the threshold c_t^j as agent j 's subjective estimate of the fundamental value; their demand function would therefore be consistent with the chartist demand introduced in Day and Huang (1990).¹⁴ However, since we recognize the opacity of such a fundamental value, we allow for a more general interpretation of the threshold level c_t^j . For example, the threshold c_t^j can also be interpreted in the context of ‘anchoring’, as was described by Tversky and Kahneman (1974). Here, investors base their demand on a certain price level that may function as an anchor. The larger the difference of the current price and the (subjective) anchor, the larger is the demand of these agents.

Putting the above demands together in Eq. (2) yields

$$P_{t+1} = a + (1 - b)P_t + \sum_{j=1}^{N_t^{CF}} \alpha_j^{CF}(P_t - P_{t-k_j}) + \sum_{j=1}^{N_t^{CT}} \alpha_j^{CT}(P_t - c_t^j) + e_t \quad (3)$$

where $a := \theta(a_R - a_S)$, $b := \theta(b_R + b_S)$, and $\alpha_j^h := \theta a_j^h b_j^h$ denotes the *price impact* of

¹⁴Furthermore, threshold traders whose extrapolation parameters b_j^{CT} is negative could be interpreted as standard fundamentalists in the sense of Day and Huang (1990).

agent j of type h . The above can further be manipulated as follows

$$P_{t+1} = [a - a_t] + [1 - b + b_t^{CT}] P_t + \sum_{k=0}^K \bar{\alpha}_{t,k}^{CF} \Delta P_{t-k} + e_t \quad (4)$$

where $a_t := \sum_{j=1}^{N_t^{CT}} \alpha_j^{CT} c_t^j$ denotes the impact of agents' thresholds on prices and $b_t^{CT} := \sum_{j=1}^{N_t^{CT}} \alpha_j^{CT}$ represents the (time-dependent) aggregate price impact of the threshold chartists.¹⁵ Finally, $\bar{\alpha}_{t,k}^{CF} = \sum_{i=k+1}^K \sum_{j=1}^{N_t^{CF}} \alpha_j^{CF} I(k_j = i)$, where $I(\cdot)$ denotes the indicator function, represents the aggregate impact of the trend-following chartists on the lagged difference ΔP_{t-k} , where K is the maximum lag used by any agent.

Finally, we observe that Eq. (4) can be rewritten to exactly correspond to the recursive ADF test specified in Eq. (1) assuming constant parameters:

$$P_t = \mu + \delta P_{t-1} + \sum_{j=1}^J \phi_j \Delta P_{t-j} + \epsilon_t. \quad (5)$$

The time-varying estimates of the parameters in Eq. (5) can therefore be mapped directly onto the variation in the impact and number of the different chartist types as they evolve through time. Specifically, an increase in the estimate of δ from successive recursive regressions directly corresponds to an increase in b_t^{CT} , indicating a larger aggregate impact of threshold (CT) traders on gold prices.

Furthermore, we see that the chosen real consumer demand and supply functions for gold would result in a stationary gold price in the absence of any speculative trading.¹⁶ In other words, real supply and demand results in gold prices that exhibit mean-reverting behaviour, consistent with standard economic theory.

¹⁵We note that the parameters a and b would be much larger in magnitude than α_j^h since the latter parameter denotes the price impact of an individual investor on gold prices and is likely very small. The aggregate affect of each agent type, however, is clearly dependent on the number of such agents active at that time, an affect that could potentially be very large.

¹⁶Recall that the price is stationary if δ in Eq. (5), or $(1 - b)$ in Eq. (3), are smaller than one.

We also note that our framework is consistent with the idea presented in Brunnermeier (2008) that if a bubble persists, and prices become too high, it becomes optimal for fundamental traders to switch to more chartist strategies and *ride* the bubble; they simply cannot afford to trade against the market. In our framework this behaviour would correspond to agents changing their parameter b_j^{CT} from a negative to a positive value in response to a persistent period of increasing prices.¹⁷

Interestingly, and perhaps counter to one's initial intuition, we also see that pure trend-following chartist activity (CFs) would not result in explosive price behaviour; as defined by $\delta > 1$ in Eq. (1). The reason for this is that such chartist activity increases the future price *proportional* to the price differences observed in the past—but never by more than that. This implies that the price process does not result in explosive characteristics. Instead, it can be seen that the explosive nature of our model above is due entirely to the threshold chartists (CTs) whose demands are price-level dependent (see also Hüsler et al., 2013). Their behaviour may result in an explosive price process since their demand is not restricted to be based only on past price changes, but on differences relative to a certain threshold or anchor. Such investors buy the asset because the current price is above a certain value, here a threshold c_t^j , and if the current price is well above the threshold the effect on the future price will be even higher. This implies that future prices can indeed increase by more than it has increased in the previous period, thereby potentially generating an explosive price process. These conclusions are in line with the existing heterogeneous agents literature where the explosive behaviour is often induced by contrarian fundamentalists (e.g. see Day and Huang, 1990) and not by pure trend-followers. When chartists are simply trend following, they are usually described as destabilizing, rather than as explosive (e.g. see Hommes and Wagener, 2009).

¹⁷The bull market between 2002 and 2012 is fully consistent with this argument (see Figure 1).

3 Empirical Results

This section consists of a presentation and discussion of the estimation results and a description of the behavioural implications of the testing framework.

Figure 1 shows the evolution of the price of gold (London AM Fix, denominated in US dollars) from January 1968 to October 2013. The figure illustrates that the gold price increased significantly from around US\$300 in the year 2000 to above US\$1,900 in 2011 and experienced a substantial correction to values around US\$1,200 in mid-2013. The magnitude of the price increase between 2000 and 2011 indicates significant speculative demand and suggests the presence of a bubble.

*** Insert Figure 1 about here ***

Figure 2 presents the test statistic and the associated p-values from our recursive test based on monthly data for the 13-year period from 2000 to 2013. The estimates are obtained with the use of the software R. The ADF test results employ the functions `adf.test` and, for robustness checks, the function `pp.test` (Phillips Perron test). The lag order is determined similar to the approach by PWY as suggested in Campbell and Perron (1991). The truncation lag parameter is found to be between three and five for the recursive samples between 2000 to 2013.

*** Insert Figure 2 about here ***

Figure 2 illustrates that statistically significant bubble characteristics are found in the period between 2002 and 2012. This bubble-like behaviour originated at the beginning of the year 2002, briefly interrupted at the end of the year 2008 and

collapsed around February 2013. Both the origination and the collapse are identified using the date stamping methodology proposed by Phillips et al. (2011).¹⁸

Mapping

We outlined in Section 2 that a bubble, when defined as an explosive price process, can be directly linked to chartists (or contrarian fundamentalists) following certain speculative trading strategies. We want to extend this interpretation further by suggesting a mapping of the p-value obtained from the recursive bubble test to the importance of chartists following strategies that lead to an explosive price process. Since the p-value is always between zero and one, it provides a simple (and almost natural) proxy for the weight or the importance of chartists relative to real demand and supply factors or investors following more stabilizing trading strategies. Hence, we interpret the p-values close to one as evidence for the dominance of real factors or investors that follow strategies that are consistent with a mean-reverting price behaviour and p-values close to zero as evidence for the dominance of threshold chartists strategies. Episodes which imply a large weight of chartists are consistent with low p-values such as the 2002 to 2012 period.

Specification issues

We performed several specification checks to examine the robustness of the findings reported above. First, we fixed the lag truncation parameter over the recursively extending sample and varied the number of fixed lags in alternative recursive estimation settings with no significant change in the estimates. One major difference is

¹⁸Estimates of the origination date r_e and the collapse date r_f are obtained as follows: $\hat{r}_e = \inf_{s \geq r_0} \{s : ADF_s > cv_{\beta_n}^{adf}(s)\}$ and $\hat{r}_f = \inf_{s \geq \hat{r}_e} \{s : ADF_s < cv_{\beta_n}^{adf}(s)\}$, where the critical values $cv_{\beta_n}^{adf}(s)$ are determined following Phillips et al. (2011).

that for large fixed lag truncation parameters the 2008 interruption disappears.

The use of daily data, and thus significantly more observations, confirms the findings obtained with monthly data. Figure 3 presents the test statistics and corresponding p-values using daily gold price observations.¹⁹ The test statistics are more volatile in general and show several brief interruptions of the 2002–2012 bull market in gold. However, qualitatively, there is no fundamental difference between the results obtained with daily and monthly data.

*** Insert Figure 3 about here ***

An alternative unit-root testing framework by Phillips and Perron, again, yields more volatile test statistics but confirms the explosive price behaviour of gold between 2002 and 2012. We also applied a forward-moving fixed window regression as an alternative to the forward recursive regression. Not surprisingly, a fixed moving window yields less stable results, i.e. more switching between periods of significant and insignificant explosive price behaviour.

Finally, as an alternative specification we tested for explosive behaviour in the log-price of gold rather than the price level. This was done by simply substituting P_t for $\ln P_t$ in Eq. (1). A value of $\delta > 1$ in such a test can be seen to indicate the existence of ‘super-exponential’ growth in the gold price level, a much stronger form of price increase, and hence bubble-like behaviour, than explosive behaviour in the price level only (see, for example, Hüsler et al., 2013). Figure 4 presents the test statistics and corresponding p-values using monthly gold price observations.²⁰

*** Insert Figure 4 about here ***

¹⁹The lag truncation parameter varies between 10 and 15.

²⁰The results at the daily frequency are, once again, very similar and therefore omitted.

We can see clearly from these results the existence of significant explosive log-price behaviour during the period 2002–2008 but insignificance after this time. This result indicates that, whilst significant explosive price behaviour in gold is observed between 2002–2012, the growth was much stronger during the 2002–2008 period, where it exhibited super-exponential growth.

The 2013 price correction

The price of gold experienced a substantial correction in 2013. The price fell from levels around US\$1,600 in March 2013 to values close to US\$1,200 in June 2013. This 25% price drop could be interpreted as a natural consequence of the bubble-like characteristics identified for the period between 2002 and 2012. If the price of gold indeed followed an explosive price process consistent with a bubble, a correction of the price of gold was perhaps an expected result.²¹

Within the context of our theoretical foundations we posit the following behavioural explanation for the recent gold price correction. Following the steady rise in gold prices over 2002–2011, the price plateaued during 2012. Given this fact, and the absence of any clear fundamental indicators, it seems likely that the threshold levels employed by speculative investors increased due to the anchoring of these thresholds to sustained higher gold prices. Such threshold adjustments would have resulted in reduced demand from these agent types, allowing other forces, such as real supply and demand, to dominate—resulting in an increased downward pressure on gold prices.

The substantial drop of the gold price in 2013 and the explosive price patterns

²¹The first version of this paper (from October 2012) did not include the 2013 price correction but reported similar results to this paper, i.e. it identified bubble-like characteristics for the 2002–2012 period suggesting that this finding implied a future price correction. The updated sample extending to October 2013 includes the 2013 price correction and therefore confirms the conclusions drawn in the first version of this paper.

identified in the period prior to this price drop supports the anticipative role of the forward recursive tests as a monitoring system for asset price time series.

4 Conclusions

This paper applied a recently developed test to the price of gold and asked the question whether historical gold prices have exhibited bubble-like characteristics. The answer is yes. We find strong evidence of explosive price behaviour. The test statistics detect periods in which the gold price followed an explosive price process between 2002 and 2012 only shortly interrupted in the immediate aftermath of the subprime crisis in late 2008. Furthermore, the test indicates that the log-price of gold also followed an explosive price process between 2002 and 2008, indicating super-exponential growth in prices during the run up to the subprime crisis. Our results suggest the presence of excessive speculation during 2002–2012 and furthermore that the high prices induced by such speculation was, at least in part, correcting during mid-2013.

We also propose a behavioural foundation for the bubble test and show that chartists, more precisely agents using certain price thresholds, dominated the gold market in recent years, meaning that the real demand for gold has declined in relative importance. This interpretation is backed by data on the demand components of gold which show that real demand has indeed significantly declined relative to investment demand (e.g. see World Gold Council, 2012).

The finding of explosive price behaviour in the gold market also implies that the role of gold as a store of value and a safe haven is under threat. Bubbles often pose a risk to financial stability but the consequences may be more severe if the asset is viewed as a “safe asset”. If the exuberance in the gold market identified in

the 2002 to 2012 period is not fully corrected by subsequent price drops there is a potential for a further decline in the price of gold. If this price drop occurs at a time when other assets fall as well, in a financial crisis for example, a falling gold price could be particularly destabilizing. However, the mere existence of bubble-like price dynamics in the gold market need not, in itself, be cause for concern. If the price were to fall without simultaneous price drops in other assets, the implications for financial stability would be less severe than a joint and extreme fall of asset prices including gold. It would possibly be even less severe than a comparable drop in equity prices or bond prices since a falling price of gold, in stark contrast to equity and bond prices, has historically signalled good news regarding financial and monetary stability. Future research could examine these issues further as well as investigating the role of an opaque fundamental value on the likelihood of the formation of explosive price processes.

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Figure 1: **Price of gold.** The graph shows the evolution of the price of gold measured at a daily frequency in US dollars from January 1968 to October 2013.

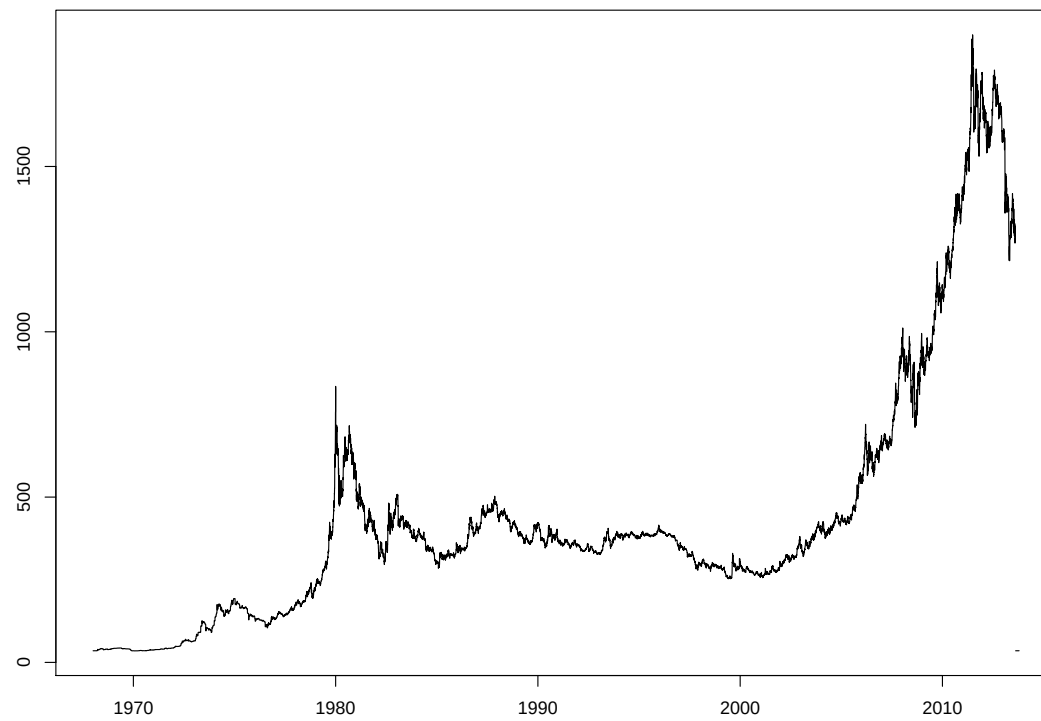


Figure 2: **Bubble test statistic and p-value. Sample: monthly prices 2000–2013.** The figure shows two plots. The top graph plots the bubble test statistic of a unit-root test with a mildly explosive process as an alternative. The bottom graph plots the associated p-value and can be used as an estimate of the amount of chartist activity in the gold market. A p-value close to zero implies that the null hypothesis of a unit root can be rejected at a significance level of $(1 - p)\%$, indicating the dominance of chartists during that period.

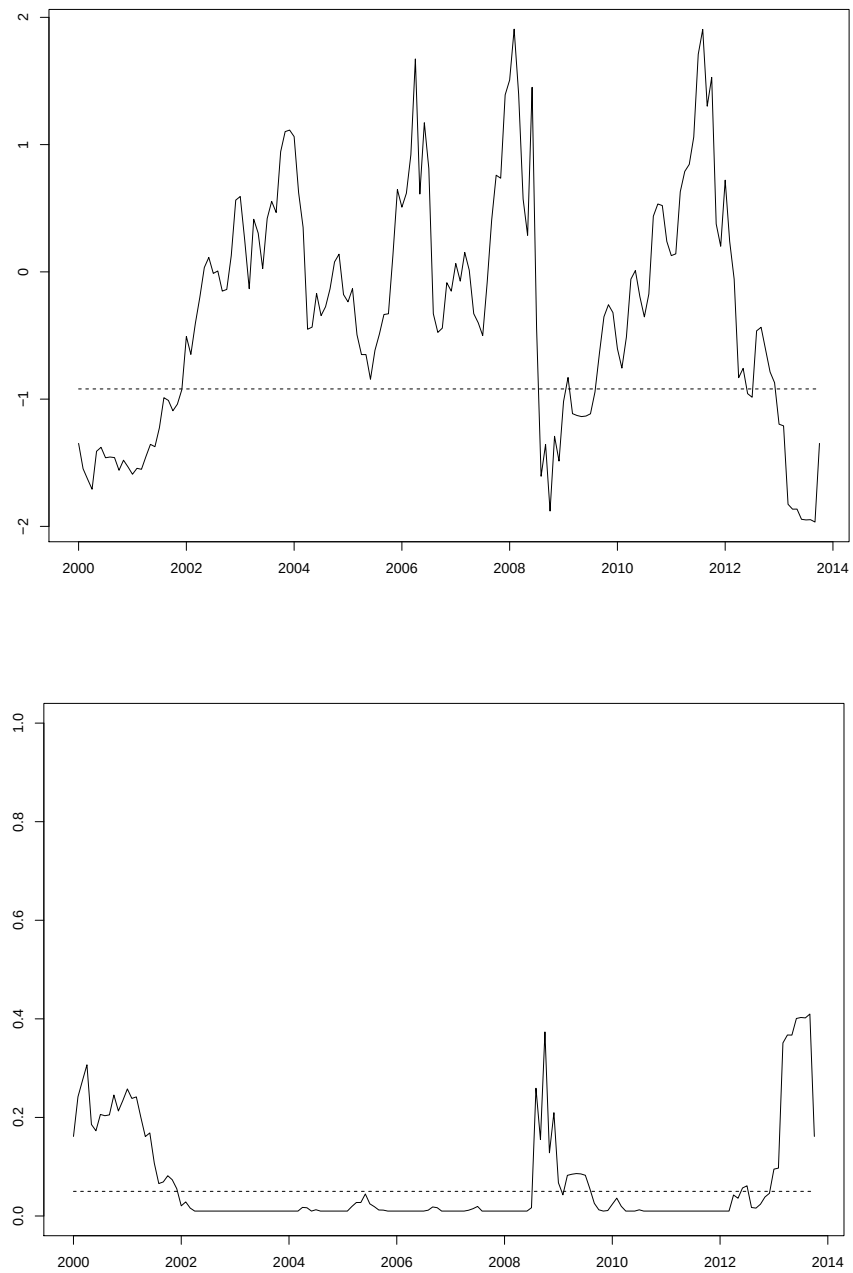


Figure 3: **Bubble test statistic and p-value. Sample: daily prices 2000–2013.** The figure shows two plots. The top graph plots the bubble test statistic of a unit-root test with a mildly explosive process as an alternative. The bottom graph plots the associated p-value and can be used as an estimate of the amount of chartist activity in the gold market. A p-value close to zero implies that the null hypothesis of a unit root can be rejected at a significance level of $(1 - p)\%$, indicating the dominance of chartists during that period.

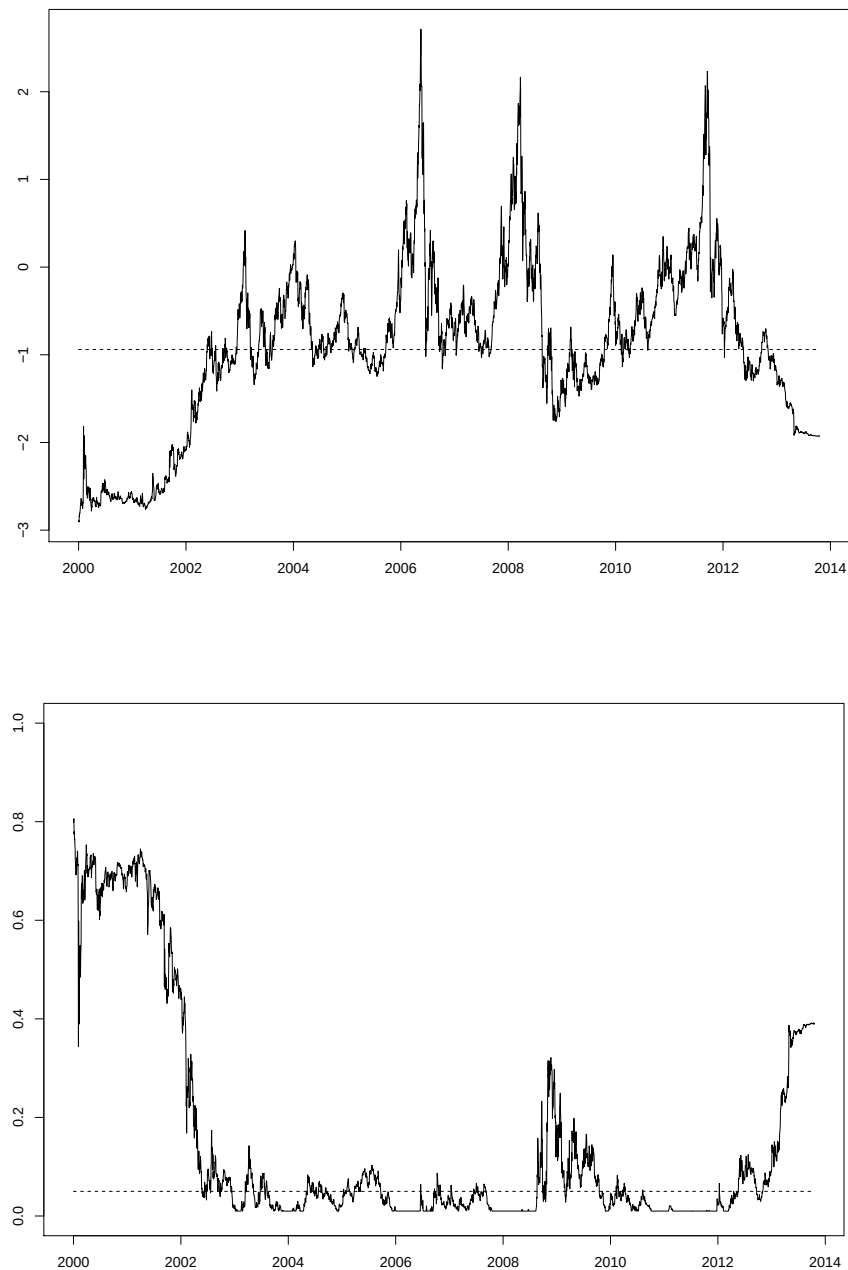


Figure 4: **Bubble test statistic and p-value (log-prices). Sample: monthly prices 2000–2013.** The figure shows two plots. The top graph plots the bubble test statistic on log-prices of a unit-root test with a mildly explosive process as an alternative. The bottom graph plots the associated p-value and can be used as an estimate of the amount of chartist activity in the gold market. A p-value close to zero implies that the null hypothesis of a unit root can be rejected at a significance level of $(1 - p)\%$.

